innovations have also led to completely drop drive cages from the cabinet. Instead, there are suspended drive bays on the motherboard plate in either of the sides. This ensures there's no obstruction at all for airflow from the front intake fans. To prevent the mess of cables affecting airflow, cabinets now have PSU shrouds or compartments. All the extra cables can be stashed in these dedicated zones, away from sight. Some cabinets have dedicated cable routes for the thick cables running behind the motherboard plate. They also have covers or shields to hide them. In conclusion, all these little changes in design have made it easier to implement better air cooling.

EXTERNAL FACTORS

Apart from the challenges present inside the cabinet, optimal airflow also depends on few external factors. The first would be the room's ambient temperature. In India, the average temperature in the major parts of the country lies between 25-27.5 degrees Celsius. During summers, the temperatures can rise way above 30 degrees. In order to keep your system cool, you could consider getting an air-conditioner or maybe even a basic room cooler. However, it's a luxury and not necessary. Another important factor is the location of your case. Placing it inside an enclosed space won't allow fresh air to enter the cabinet. It would be best to place it in the open either on your desk or directly on the floor. Also, avoid keeping your cabinet over a carpet or rug since it obstructs the PSU fan. If your room has other machines that generate heat, it would be best to move your case to a cooler corner. Essentially, the idea is to keep an unending supply of fresh and cool air around the cabinet.

DIFFERENT FAN MOUNTING LOCATIONS

In the most basic cabinets, you will find provision to install two front intake fans, one exhaust fan on the rear and two more exhaust fans on the top. This is generally the case with cabinets in the lower price range. If you go a little higher, you will come across cabinets additionally allowing three front intakes and three exhaust fans on top. Some cabinets might even allow you to add one or two fans at the bottom and side panel. Going up the ladder will only improve the number of supported fans. In the premium range, you will come across enthusiast cabinets such as the Thermaltake WP200 that can accommodate a mind-boggling 40 fans in the cabinet. As we said earlier, more fans doesn't necessarily mean a cooler system. In fact, a four or five fan setup is sufficient to keep things cool.

Entry-level cabinets have fan mounts on the front, rear and top. Some of them might even include additional mounts at the bottom or on the side panels. Conventionally, the front and bottom fans are used for air intake while the rear and top fans are used for exhaust. The front panel usually has the provision to accommodate the most number of fans in basic cabinets. Every cabinet is designed with a particular airflow pattern. Air will flow from the front to the rear and from the bottom to the top. Since hot air rises, it makes perfect sense to use the top fans as exhaust. If your cabinet has fans on the side panels, your configuration will be dependent on the fans you use. We will now explore the different kind of fans you can install in your cabinet.

Conventional flow of air inside PC cabinets

CASE FAN TERMINOLOGY

They are generally available in sizes of 92mm, 120mm, 140mm and 200mm. Cabinets nowadays come with 120mm or 140mm fan mounts, the former being more popular. Before going into details, there are a few fan terms we would like to explain. The first is fan speed measured in RPM that indicates the number of times the fan will rotate in one minute. Then we move on to CFM or cubic feet per minute. This tells you how much air the fan is capable of moving. The higher the CFM value, the better the fan's performance. Next is static pressure measured in mmH2O. The static pressure value indicates how much force the fan is able to exert on an object. These fans are used when there's obstruction in front of fan such as a radiator, heat sink or HDD cage. You will find fans that will prioritise airflow or static pressure, and accordingly you can buy them based on your configuration. The final term is noise level. Although this isn't important, it's better to know that higher fan speed will result in higher noise levels. Also, some manufacturers sell quiet versions on both airflow and static pressure fans. With the important terms out of the way, let's get into the final segment of this guide.

SETTING IT ALL UP

Before taking a plunge into fans, let's start off with cable management. Mid-tier cabinets are generous enough to accommodate extra cables. You can simply hide them without worrying about